# Promptly Health — Interview Prep

Preparação: João Fonseca (Data Lead) e Max Gleichauf (BD)

# Promptly Health — Interview Preparation Dossier

**Role: Junior Clinical Data Engineering / RWE Analytics**

**Panel: João Fonseca (Data Lead) · Max Gleichauf (Business Development & Strategic Partnerships)**

**Prepared from: a full read of `FHIR-to-OMOP`, `CDISC-to-OMOP`, `Clinical-data-to-CDISC` + Promptly's public technical positions**

---

## 0. HOW TO USE THIS DOCUMENT

- **Sections 1 and 2** are question-by-question. Each has: what he is really testing → the trap → the answer structure → a spoken script.
- **Section 3** is the part that will save you. Read it twice. A senior data lead does not reject juniors for not knowing; he rejects them for **not knowing what they don't know**.
- **Section 4** has your closing questions. Ask them verbatim-ish. They are designed to be unanswerable by someone who only skimmed your README.
- **The Appendix** is your one-page cheat sheet. Print it.

**Your single strategic thesis (memorise this; everything ladders back to it):**

> "I build mapping systems where the AI is allowed to be wrong, but is never allowed to be silently wrong. Every concept that enters a clinical table carries provenance: which run, which model digest, which vocabulary slice, which index signature, which two humans approved it, and which adjudicator signed. In a federated network that is the whole product — because if a partner hospital cannot prove how a row got there, the row is not evidence."

That sentence is simultaneously the answer to the DuckDB question, the abstention question, and Max's governance question. It is also, almost word for word, Promptly's own problem.

---

# 1. PROBING TECHNICAL QUESTIONS — JOÃO FONSECA

## What you must understand about João before he speaks

From his own published interview (Promptly Health, 03/01/2024), in his own words:

- He came to OMOP from an **MSc thesis on AI mortality prediction** where he hit the classic wall: "conducting comparisons without identical data sets." He is therefore **wired to distrust a result he cannot reproduce on a fixed input.**
- He uses the words **"reusability" and "scalability"** as his explicit justification for OMOP. Not "interoperability." Reuse across clients, and scale across clients.
- He says Promptly's harmonisation bottleneck is handled with **USAGI + clinical specialists refining it**, and that "efforts are underway to develop a more efficient AI-based tool leveraging vast medical concepts in the source database, retraining the model to improve vocabulary mappings and reduce manual labour."
- And critically, on new data sources: "The remaining task involves analyzing the client-specific data model and establishing the 'load' aspect of the pipeline."

**Read that last line again.** João's mental model is: the analytics and the transform are solved and standardised; the variable cost is per-client source analysis. He spends his life on the messy front end.

**Your entire portfolio is an answer to that sentence.** You have built, three times over, the thing he says is the remaining work — with the AI layer he says they are trying to build. When he reads your repos he is not seeing "a junior who learned OMOP." He is seeing someone who independently built a governed version of the exact tool his team is trying to ship. Say that out loud, once, without arrogance.

The Promptly poster (OHDSI, 23/10/2024, "Improving vocabulary mapping to OMOP-CDM concepts: an AI-based automated and collaborative mapping platform into the ETL pipeline") describes an **alpha** where "clinicians search and select OMOP-CDM concepts collaboratively" and mappings are **exported** for database use. You built the same loop — but with two blinded reviewers, a separate adjudicator, rejection-as-policy, and fail-closed parsing. **You are one design iteration ahead of their alpha on governance.** Do not say that as a boast. Say it as an offer.

---

## 1.1 — "Why DuckDB for local transformation? Why not Postgres or Spark?"

### What he is actually testing

Whether you chose a technology or inherited one. Whether you understand that in a federated network the ETL engine is not a database decision — **it is a deployment, governance and reproducibility decision.** And whether you know DuckDB's one real limitation (single-writer).

### The trap

Answering "it's fast, it's embedded, it's easy." That is a blog-post answer. He will then ask "what happens when two processes write?" and you will be exposed.

### Answer structure (4 beats, then stop)

- **Reframe to the constraint.** The node is inside a hospital. You do not get a DBA, you do not get an open port, and you certainly do not get a shared cluster.
- **Postgres: rejected on operational governance, not on merit.**
- **Spark: rejected on unit economics of the transform, not on capability.**
- **Concede the real limitation** (single writer) and show how your design handles it — because that is the same problem they will hit.

### Spoken script (~55 seconds)

> "I want to separate two things people usually merge: the engine that builds the CDM, and the substrate that serves it. My choice only concerns the build.
>
> The build runs inside a partner hospital. So the decision isn't really Postgres-versus-DuckDB on performance — it's: what can I install in a locked-down environment without a DBA, without opening a network port, and without a service account that has to survive an IT governance review? Postgres is a service. It needs a daemon, a port, a backup policy, a credential rotation story, and someone to own it at 400 hospitals. DuckDB is a library and a file. My ETL is a container that produces one artefact.
>
> Spark I ruled out on unit economics. An OMOP ETL is overwhelmingly joins, window functions and deterministic tie-breaking — I'm collapsing eras with a 30-day persistence window and exploding drug products through `CONCEPT_ANCESTOR`. That's not shuffle-bound work. Spark buys me parallelism I don't need and charges me a JVM, a cluster, and non-determinism I'd then have to fight, because I need byte-reproducible runs.

> The reason I care about reproducibility is the reason I picked DuckDB: **the published database is a single file I can SHA-256.** Every run records the git commit, an input manifest hash, the vocabulary version, the Chroma index signature, the model digest and the generation parameters. In a federated network where nobody centralises the data, that hash is the audit trail. Row-level data doesn't travel — but a fingerprint plus a run manifest does, and it lets a coordination centre ask 'are you running the same pipeline I think you're running?'
>
> The honest limitation: DuckDB is single-writer. I don't pretend otherwise. I handle it with a non-blocking advisory `flock` on a `.publish.lock` sidecar, and — more importantly — by never writing to the published file at all. Every run builds into an isolated staging database and the published file is swapped only after all gates pass. That's also why my DQD runner shards the two memory-heavy check families into separate R processes: I'm working with the single-writer grain, not against it."

### The follow-up he will fire, and your answer

**"So what's your story for serving queries at scale — Achilles, Atlas?"**

> "I haven't built that layer, and I'd rather say so than hand-wave. My pipeline terminates at a validated v5.4 CDM plus DQD evidence. The serving layer — Achilles characterisation, Atlas/WebAPI — is exactly the part where I'd expect Postgres or a warehouse to take over, because that's where concurrency and multi-user access actually matter. My view is that engine choice should follow the access pattern: single-writer, reproducible, hashable artefact at the node; concurrent, networked serving above it. If Promptly is running Athena/Presto or a warehouse over the federated layer, I'd want to understand where that boundary sits — which brings me to a question I actually have for you later."

**Why that lands:** it converts a gap into a question, and it proves you think in boundaries, not in favourite tools.

---

## 1.2 — "Walk me through how your LOINC reranker handles a messy, abbreviated hospital lab description."

### What he is actually testing

This is the highest-value question in the interview. He wants to know whether you understand that **the reranker must never impute a missing axis.** LOINC is multi-axial: Component, Property, Timing, System (specimen), Scale, Method. A short LIS label like `"CREA"` or `"Hb sangue 13,6 g/dL"` carries maybe two of those six. A junior's instinct is to fill in the blanks. That is how you map urine creatinine onto serum creatinine and poison a renal-function phenotype across an entire network.

He also wants to see whether you know that **retrieval recall is your real ceiling**, not the reranker.

### The trap

Describing the reranker as if it improves recall. It cannot. It only reorders what Chroma already returned. If the right LOINC code is not in the candidate pool, reranking is theatre.

### The mechanics you must have at your fingertips

From `src/mapping/loinc_reranking.py`:

- Versioned as `loinc-axes-v1`, and the version is recorded in mapping audit metadata.
- Measurement retrieves a **wider pool of 20** (`LOINC_RETRIEVAL_POOL_SIZE`) and reranks down to **top 5**. Non-measurement domains retrieve `top_k` directly — the reranker is Measurement-only.
- Three axes are handled: **specimen, property, method**. Patterns are explicit regexes over a casefolded, bracket-stripped, whitespace-normalised string.

- Scoring: start from Chroma cosine distance converted to a bounded similarity (`1 - distance`, clamped to `[0,1]`); **+0.08 for an axis match**, **−0.35 for an axis conflict**.
- **Asymmetry is the whole design.** A match is worth +0.08; a conflict costs −0.35. Reward is a nudge, penalty is a veto.
- **A conflict only fires if the candidate actually has that axis populated.** If the candidate's axis is empty, nothing happens — no reward, no penalty. That is the "never impute" rule, encoded as an `elif candidate_values:`.
- Tie-break is `-rerank_score, original_rank` → fully deterministic.
- **Original Chroma distances are preserved** in the output for governed scoring. The reranker reorders; it does not rewrite evidence.
- One disambiguation rule: if `arterial_blood` matched, `blood` is discarded — a deliberate hierarchy, not an accident.
- Non-identifying context (`Units: ...; Observed value kind: numeric`) is appended to the **rerank input** and shown to the model — **but individual patient values are never sent to the model.** Units and value kind may inform reranking; the number itself does not.

Also relevant, from `src/mapping/retrieval_normalization.py`: a **versioned LIS alias lexicon** (`lis-aliases-v1`) that expands exact governed aliases (`CREA` → `"Creatinine [Mass/volume] in Serum or Plasma"`) before retrieval. It is retrieval-time normalisation only — it never assigns a concept. And note the safety valve: the lexicon is scoped `generic-synthetic-development`, so **it is deliberately inert when `data_classification` is not `SYNTHETIC`.** It cannot silently alter behaviour on real PHI.

## Spoken script (~70 seconds)

> "There are two separate problems and I built two separate mechanisms, because conflating them is where people go wrong.
>
> **First, abbreviation.** `CREA` isn't a retrieval problem, it's a vocabulary problem — the embedding model has nothing to grab. So I use a versioned, hash-pinned alias lexicon that does exact-match expansion before retrieval: `CREA` becomes `Creatinine [Mass/volume] in Serum or Plasma`. Critically, that's a **retrieval-time normalisation only — it assigns no concept.** And it's gated: the lexicon is scoped `generic-synthetic-development`, so on a PHI-classified run it's deliberately inert. I don't want a synthetic-dev convenience silently changing behaviour on real patient data.
>
> **Second, disambiguation.** For Measurement I retrieve a wider pool of 20 and rerank to 5. The reranker extracts only axes that are **explicitly present in the source text**, and then does something deliberately asymmetric: an axis match is worth **plus 0.08**; an axis conflict costs **minus 0.35**. The reward is a nudge, the penalty is a veto.
>
> The rule that matters most is the one that doesn't fire. A conflict is only scored if the candidate has that axis populated. So for `"glicose 104, espécime e método ausentes"` — glucose, but specimen and method genuinely absent — a serum candidate is **not penalised** for lacking urine, and a urine candidate is **not rewarded** for having it. Missing means unknown, and unknown means the axis is silent. **The reranker never imputes a missing axis.** If it did, I'd be laundering an assumption into a clinical concept, and the failure would be invisible: it would look like a perfect mapping right up until someone builds a phenotype on it.
>
> And the limitation I'd state before you ask: **reranking cannot improve recall.** It only reorders what Chroma returned. On my held-out set, Measurement top-k recall was the weakest domain in the benchmark — which tells me the honest next investment isn't a smarter reranker, it's a better retrieval layer, probably domain-tuned or multilingual embeddings."

**That last paragraph is the one that gets you the job.** You are quoting a number that makes you look worse, unprompted, because it is true. João's entire published worldview is "comparisons without identical data sets" are the cardinal sin. Volunteering a limitation is the single highest-signal thing a junior can do.

### The follow-up he will fire

**"Your specimen patterns are English regexes — `urine`, `serum`, `plasma`. Portuguese hospitals write `urina`, `soro`, `plasma`. What happens?"**

Do not improvise. Answer:

> "They don't match, and I'd rather that than a false match. The regexes are English-only, so a Portuguese specimen token is simply invisible to the reranker — the axis goes silent and the candidate survives on embedding similarity alone. That's a **degraded but safe** state: I lose disambiguation power, I don't inject error. The correct fix is a versioned, locale-scoped axis lexicon — `loinc-axes-v2-pt` — with the same asymmetry and the same hash pinning, validated against held-out Portuguese cases before it's allowed to score. I would not ship language coverage without that, because a specimen lexicon has a false-friend problem: `soro` is serum, but contextually `soro` in a Portuguese ward can mean an IV infusion, and that ambiguity is exactly the kind of thing that should reach a human."

---

## 1.3 — "Your abstention is fail-closed. A client wants 100% mapping coverage. What do you tell them?"

### What he is actually testing

This is a **commercial maturity test disguised as a technical one.** Promptly sells to pharma and hospitals. Someone will demand a coverage number. João needs to know whether you will cave under commercial pressure and quietly lower a threshold — because if you will, you will eventually ship a wrong mapping into a regulatory-grade study.

He is also testing whether you understand that **coverage is not one number.**

### The trap

Two traps. (a) Defending abstention on purity grounds — you sound academic and unbusinesslike. (b) Offering to "lower the threshold" — you sound dangerous.

### The honest numbers you must be able to quote

From `benchmarks/dirty_hospital/phase5_held_out_summary.json` — held-out split, 40 cases, 30 expected MAP / 10 expected ABSTAIN, model `qwen2.5-coder:7b`, `temperature=0.0`, `seed=0`, `num_predict=512`, prompt `mapping-json-v2`, vocabulary `v5.0 27-FEB-26`, zero contract failures:

| Arm | Accepted | Accepted precision | Coverage | Mappable recall | Abstain accuracy | False maps |
|---|---|---|---|---|---|---|
| Deterministic code-only | 10 | **1.00** | 0.25 | 0.333 | 1.00 | **0** |
| Retrieval + LLM (qwen) | 10 | **1.00** | 0.25 | 0.333 | 1.00 | **0** |

And the calibration curve from the embedding-retrieval arm makes the trade-off explicit: at threshold 0.0 → **precision 0.35, coverage 1.00, 10 false maps**. At 0.5 → precision 0.45, coverage 0.775. You are buying precision by spending coverage, on a visible, auditable curve.

**Say this out loud: on that held-out set the LLM arm did not beat the deterministic arm.** Identical accepted set, identical precision, identical coverage — and it cost 97 seconds versus 0.42 seconds.

### Why that is strength, not weakness

Anyone can get 100% coverage. Set the threshold to zero and accept the top candidate every time. You get coverage 1.0 and **precision 0.35** — which means **ten confident, wrong mappings, each one**

**indistinguishable from a correct one once it lands in `condition_concept_id`.** That is not coverage. That is manufactured evidence.

### Spoken script (~70 seconds)

> "I'd reframe the ask, because 'coverage' is being used as if it were one number and it isn't. I'd ask which of three things they actually want:
>
> **One — coverage of source codes.** That I can drive very high, because it's a function of vocabulary breadth and reuse, not of model risk. My rejection policy is reusable: once a human approves a mapping it applies on the next run, and once they reject one it becomes active policy that suppresses identical future proposals and stops the model being called at all. Coverage compounds across runs and across clients. That is the lever, and it's the one João's own 'reusability and scalability' argument is about.
>
> **Two — coverage of clinical events. **Different question, and the one I actually want. An unmapped event isn't dropped — it's published as `concept_id = 0` with its source value, source concept and unit provenance intact.** Nothing disappears silently.** The event is still countable; it just isn't yet semantically interoperable.
>
> **Three — coverage of analysable phenotypes.** That's the only one that should drive study feasibility, and it's the only one that should be reported per-study, not per-database. A lab panel for a renal outcome needs different coverage than a comorbidity score.
>
> Then I'd show them the curve. On my held-out set, dropping the threshold to zero gives coverage 1.0 and **precision 0.35 — ten confident wrong mappings.** And I'd tell them the unflattering part first: on that set, the LLM arm and the deterministic arm produced the same accepted set, same precision, same coverage — the LLM cost ninety-seven seconds and bought nothing measurable on that slice.
>
> So my answer to the client is: **you can have any coverage number you like; you can't choose the precision that comes with it.** What I'll commit to is a coverage figure at a stated precision floor, reported per domain, with the abstention queue as a visible, sizeable backlog — and I'd rather hand them a worklist of 400 terms for a clinical specialist than a dashboard claiming 100% that quietly contains thirty wrong concepts. In a federated network a wrong mapping doesn't stay local. It propagates into every study that runs on that node."

**The killer close:** "In a federated network a wrong mapping doesn't stay local." That is Promptly's own architecture argument turned into a governance argument.

---

## 1.4 — "How do you validate DQD failures? Walk me through Plausibility, Conformance and Completeness."

### What he is actually testing

Whether you treat DQD as **a gate or a dashboard**. Most candidates run DQD, see 96% pass, and move on. João runs it as a gate (`include_dqd: true` is mandatory in hospital profile). He wants to know what you do with the 4% that fails — and whether you know the **difference between an error and a failure**.

### The mechanics you must have at your fingertips

From `src/quality/validate_dqd.py` and `quality/dqd_policy.json`:

- The gate requires `policy["status"] == "approved"`. **An unapproved policy hard-fails.** The policy is a reviewed artefact with `reviewed_by`, `reviewed_on`.

- It distinguishes **errors** (`isError = 1`: the check could not run) from **failures** (`failed = 1`: the check ran and the data violated a threshold).
- `max_errors: 0`**.** Zero tolerance. An error means you do not know — it is not a data quality signal, it is a blindness.
- `max_failed_checks: 0`**, with an explicit, human-reviewed allowance register** (`allowed_failure_ids`).
- **Bidirectional enforcement.** Every failed check must be in the register (`unjustified`), and every entry in the register must correspond to a currently failing check (`stale`). **Stale allowances fail the build.** This is the detail that will impress him: most teams build an allowlist that silently rots into a permanent excuse.
- Each allowance needs a **non-empty reason** and **numeric caps** (`max_violated_rows`, `max_pct_violated_rows`). A waiver without a ceiling is not a waiver, it's a blank cheque.
- The runner shards the **memory-heavy** `plausibleValueHigh` **and** `measureValueCompleteness` **families into isolated R processes** per table, verifies every shard is disjoint, and merges without excluding or weakening any check. `OBSERVATION` is split further into `field_measure` and `field_high`.
- Shards are **contract-verified**: schema version, mode, table, database MD5, thresholds MD5, DQD version — resuming against changed inputs is refused, not silently accepted.
- One honest engineering wart you should know and can mention: SqlRender translates `DATEADD(dd,1,GETDATE())` into unstable DuckDB `TO_DAYS` SQL, so the runner rewrites those 55 rows to `CURRENT_DATE + INTERVAL 1 DAY` — and **hard-stops with "dialect review is required" if the count ever stops being exactly 55.** That is a tripwire against a silent vocabulary/toolchain drift.
- Versions: `DataQualityDashboard 2.8.9`, `DatabaseConnector 7.2.0`, `SqlRender 1.19.6`, CDM 5.4, `cdmDatabaseSchema = "main"`, `numThreads = 1`.

## Spoken script (~75 seconds)

> "I treat DQD as a **gate, not a dashboard**, and the first distinction I enforce is error versus failure. An error is `isError = 1` — the check couldn't execute. That's not a data-quality signal, that's **blindness**, so my error budget is hard zero. A failure is `failed = 1` — the check ran and my data violated a threshold. Those I handle category by category.
>
> **Conformance failures are usually my bug.** Wrong domain routing, a non-standard concept where a standard one was required, a broken foreign key, a `concept_id = 0` where the model required non-null, a type concept outside the allowed set. I don't write allowances for conformance. I fix the ETL, because a conformance failure means I violated the CDM contract, and the contract isn't negotiable.
>
> **Plausibility failures are usually a real signal, or a real edge case.** A plausible-value-high on a lab, a future date, an implausible age. Here I want to know whether it's a unit error, a date-parse error, or a genuinely extreme patient. The classic one is a future date: my DQD runner preserves every single `plausibleValueHigh` check — it shards them into isolated processes rather than dropping them — because 'this check is expensive' is never a reason to stop asking the question.
>
> **Completeness failures are the ones I expect to negotiate, because they're usually a property of the source, not of my pipeline.** If a hospital's LIS never captured a field, no ETL can invent it. But I won't record that as a silent pass — it goes into a reviewed allowance register with a **named reason and numeric caps on both violated rows and violated percentage.** And the part that matters: the register is enforced **in both directions**. If a check stops failing, a stale allowance **fails my build.** Otherwise the allowlist quietly becomes a permanent excuse, and two years later nobody remembers why three checks are exempt.
>
> Last thing: the whole gate refuses to run unless the policy itself is marked `approved` by a named reviewer with a date. **The budget is a reviewed artefact, not a config value.** And I'd add the caveat I care about most — DQD tells me the CDM is internally coherent. It does **not** tell me the mappings are clinically right. That's a separate claim, and it needs clinical validation, which is why my benchmark is explicitly labelled `PROVISIONAL_TECHNICAL` with `clinical_validation_required:`

`true`."

**If he asks "what's a DQD check that passes and still misleads you?"** — have this ready:

"`measureValueCompleteness` on a column that's populated with a default. Or a conformance check on `concept_id != 0` — that passes as soon as I map everything to something, including something wrong. **DQD is structurally unable to detect a plausible-but-wrong mapping.** The only defence is provenance and human review, which is why I measure accepted precision on a held-out benchmark rather than trusting the pass rate."

---

## 1.5 — Six more probes João will very likely fire

### (a) "How do you prevent SQL fan-out when a source code maps to multiple concepts?"

"`QUALIFY ROW_NUMBER() = 1` with a deterministic ordering, after filtering to current Standard concepts via the `Maps To` relationship and the correct domain. But I treat fan-out as a symptom, not a nuisance: if one source code maps to many standard concepts, that's usually a granularity mismatch — the source is more granular or less granular than the target. Silently picking row one hides that. So the tie-break is deterministic and auditable, but I also record the candidate set, so a reviewer can see what was discarded. And I keep `*_source_concept_id` and `*_source_value`, so the mapping is always reversible — I never lose the ability to go back to what the hospital actually said."

### (b) "Your pseudonymization key — what happens if it changes between runs?"

"It fails closed. Every successful database records a singleton pseudonymization-key fingerprint, and a hospital run **aborts** if the institution-managed key identity changes between runs. This is the thing I'd flag as most important for a federated network and least discussed: if a node silently re-pseudonymizes, every `person_id` changes, every longitudinal link breaks, and — worse — the same patient can appear twice across runs and be double-counted in a federated aggregate. So historical identifiers require an explicitly governed migration, never a silent re-key. In a federated study that's not an inconvenience, it's a data-integrity event."

### (c) "How do you make an LLM mapping pipeline deterministic?"

"Three layers. **Input determinism**: the model only ever sees candidate IDs retrieved from a hash-pinned, version-signatured Chroma index, and it may only select an ID that was actually supplied — invented IDs fail closed. **Generation determinism**: `temperature=0.0`, `seed=0`, `num_predict=512`, and I record the model digest, not just the model name, so a silently updated `qwen2.5-coder:7b` doesn't quietly become a different system. **Contract determinism**: strict JSON schema, `additionalProperties: false`, exact key-set equality, and any violation becomes an explicit technical ABSTAIN with reason `INVALID_LLM_RESPONSE` — never a retry, never a best-effort parse.

The honest caveat: **that's determinism of the pipeline, not determinism of the model.** A different GPU, a different Ollama build or a different quantisation can change output with the same seed. So I never treat the LLM output as the source of truth — it's a proposal that two humans and an adjudicator must approve before it can enter a clinical table."

### (d) "Why the dependency-free `clinical_mapping_core` boundary?"

"Because the contract is the thing that has to be provable. `src/clinical_mapping_core/` holds the request/decision/provenance value objects, the fail-closed parser and the prompt renderer, with **zero dependency on DuckDB, Chroma, Ollama or FHIR.** That means the safety-critical logic — the part that decides whether an LLM response becomes a clinical mapping — is testable in

milliseconds with no database, no model and no network, and it can't be quietly coupled to an adapter's convenience.

It's also what let me add a second source adapter. The hospital-CSV path converts into the same typed `MappingRequest`, so it inherits the same fail-closed guarantees instead of reimplementing them. That's the design property Promptly needs most: **a new client's data model shouldn't mean a new governance model.**"

### (e) "Two blinded reviews plus a separate adjudicator — isn't that overkill? It'll bottleneck."

"It's overkill for a single-site research database and it's the minimum for a federated network. The constraint isn't statistical, it's **separation of duties** — governed `actor_id` values, not typed names, so the same person cannot be both reviewers, and neither can be the adjudicator. Aliases handle accents and verified name variants without weakening that.

On throughput: the bottleneck is bounded, because rejections become reusable policy. A rejected term **suppresses identical future proposals and prevents the model call entirely**. So the human cost concentrates on genuinely novel terms and decays as the mapping set matures — which is the same reuse curve João describes for vocabulary mapping across clients."

### (f) "Where does your `concept_id = 0` policy bite?"

"`0` is not a failure state, it's a declared unknown, and I'm strict about the difference. A supplied-but-unsupported unit maps to `0`. A genuinely absent unit stays `NULL`. Those mean different things and I refuse to collapse them: `0` means 'I saw a unit and couldn't standardise it', `NULL` means 'no unit was recorded'. If I collapse them, a completeness problem becomes indistinguishable from a mapping problem. Same principle on clinical events — unmapped events publish as `0` with full source provenance rather than being dropped, and domain conflicts go to an auditable **quarantine** instead of vanishing."

---

# 2. STRATEGIC & RWE QUESTIONS — MAX GLEICHAUF

## What you must understand about Max

Max is not a statistician and will not test your maths. He is testing **whether you are presentable to a pharmaceutical client and to a consortium partner.** He sits in rooms where someone says "can your network answer this question?" and he has to know, in real time, whether the answer is credible.

His world: **Promptly Harmonize** (containerised OMOP ETL + vocabulary mapping, "research-ready datasets in weeks, not months"), **Explore by Promptly** (natural-language and live cohort discovery, with **POLLY** — a governed library of clinical definitions that teams Define → Approve → Trace), the **Evidence Network** ("moving insights, not data"), **EHDS**, **DARWIN EU**, and the **FRAME** project with DARE UK — publicly described as "Federated Research on Anti-hypertensives Maternity Emulation Trial", £446,795, part of the DARE UK Research Exemplars portfolio, which tests federated analytics across multiple Trusted Research Environments.

**The insight to exploit:** POLLY's entire premise is that "what counts as a readmission? a new diagnosis? uncontrolled diabetes? Every team answers these slightly differently, and the answer changes the result." That is **a vocabulary-mapping problem wearing a different hat.** You have spent a year building governance for exactly that class of disagreement. Connect those two dots for him and you stop being a candidate and start being a colleague.

---

# 2.1 — "How does bridging CDISC to OMOP support Target Trial Emulation and synthetic control arms?"

## What he is testing

Whether you understand why a pharma client would pay for this. And whether you know the difference between "external control arm" and "synthetic control arm" — most candidates use them interchangeably.

## The four beats

**1. The trial gives you the protocol; RWD gives you the comparator.**

> "A randomised trial is a specification — eligibility, treatment strategies, time zero, follow-up, outcome, causal contrast, analysis plan. An RCT is expensive mainly because of that specification and the people recruited to satisfy it. RWD is cheap on people and expensive on specification. TTE is the discipline of importing the specification onto real-world data. Synthetic and external control arms are what you get when you only need to import half of it — the comparator arm."

**2. The bridge is an eligibility-criteria compiler — and CDISC is the source of truth.**

> "Here's the part I think is underrated. Trial eligibility criteria live in the protocol and get operationalised into SDTM: inclusion/exclusion flags, `IE` domain, `DM` reference dates `RFSTDTC`/`RFENDTC`, population flags in `ADSL` — `ITTFL`, `SAFFL`. If I map SDTM to OMOP and preserve that structure, I'm not just moving rows between models — I'm turning **prose eligibility criteria into executable cohort definitions** that can run against observational data."

**3. Synthetic vs external — know the difference.**

> "An **external control arm** uses real patients: different people, same disease, from routine care. A **synthetic control arm** typically means the comparator outcomes are modelled or generated — predicted from a model trained on historical controls, sometimes with no real comparator patients at all. They have different regulatory appetites. EMA and FDA have both accepted external controls in single-arm settings; fully synthetic arms are a harder sell and usually need explicit pre-specification and sensitivity analysis. I'd never let a client use those words interchangeably in a submission."

**4. The two failure modes that kill these studies — and why they're data engineering, not statistics.**

This is where you win the question. Name them:

> "**Immortal time bias**, from time-zero misalignment — eligibility and treatment assignment and follow-up start must all be anchored at one index date. And **prevalent-user bias**, from mixing patients who were already on treatment with new initiators.
>
> Both are created in the ETL, not in the analysis. If my observation periods are derived lazily — say, from the first event to the last event — then every patient is immortal by construction and no amount of propensity scoring rescues it. That's why in `CDISC-to-OMOP` I derive `OBSERVATION_PERIOD` from the strict CDISC reference dates in `DM`, not from observed events, and I type it explicitly as `Standard algorithm` versus `Case Report Form` for the clinical records. If I got that wrong, the harmonisation would look flawless and the hazard ratio would be wrong."

**5. The close — say what you cannot do.**

> "What I'd be honest with a client about: harmonising trial data to OMOP makes a comparator findable and computable. It does not make it exchangeable. Residual confounding — frailty, symptom burden, clinician judgement — isn't in either data source, and no CDM fixes that. The deliverable I'd promise is a transparent, pre-specified, reproducible emulation with negative controls and a documented mapping provenance, so that a regulator can see exactly how every

eligibility criterion was operationalised. Not a claim that we recreated randomisation."

### Optional killer line if he warms up

"And the reason I built record-level lineage in `source_record_lineage` — SDTM domain, subject, sequence, source file, verbatim value, publication `run_id` — is that in a regulatory context, 'we mapped it' is not an answer. The answer is 'here is the exact SDTM row this OMOP record came from, and here is the run that produced it.' That's what turns a harmonised database into something an assessor can audit."

---

## 2.2 — "Talk me through your pivot from Biochemistry and customer support into Clinical Data Engineering."

### What he is testing

**Risk.** Max's real question is: "will this person be stable, and can I put them in front of a hospital CIO or a pharma client?" He is not looking for a redemption arc. He is looking for **agency** — did you drift, or did you choose?

Do not apologise. Do not over-explain. Do not disparage the old roles.

### The structure: name it, credit it, convert it, land it

**Name it (one sentence, no flinch).**

"It's a deliberate pivot, not a drift. I finished a BSc in Biochemistry, worked in operations and customer support for three years, and in early 2024 decided the thing I wanted to build was clinical data infrastructure. Everything since has been pointed at that."

**Credit it — this is the move most candidates miss. Do not treat the three years as dead time.**

"And I want to be precise, because it's tempting to treat those three years as a gap and they weren't. Three years in customer support and administration gave me two things I now use every week.

**The first is stakeholder empathy.** I spent three years translating between people who had a problem and systems that had a process. That is exactly what vocabulary mapping is: a hospital LIS writes `CREA`, OMOP wants `Creatinine [Mass/volume] in Serum or Plasma`, and someone has to own the translation without losing meaning and without inventing it.

**The second is operational precision.** In support, a missed detail is a ticket that reopens. In an ETL, a missed detail is a wrong concept in a regulatory study. The discipline transfers directly — which is why my pipelines fail closed rather than guessing. A system that quietly guesses is just a support queue you can't see."

**Convert the science (don't let the BSc be decoration).**

"The Biochemistry degree is doing more work than it looks like. It's why I'm comfortable being the person in the room who says 'that lab value is implausible.' It's why I built a reranker that penalises a **urine-versus-serum** mismatch four times harder than it rewards a match — because I know what a specimen error does to a result. And it's why I treat `concept_id = 0` and `NULL` as different statements. Most people entering this field from pure CS treat clinical fields as strings. I don't."

**Land it (evidence, not aspiration).**

"The pivot is already executed, not planned. Three production-grade repositories, an end-to-end FHIR-to-OMOP pipeline with DQD validation, a CDISC-to-OMOP ETL with record-level lineage, a SAS/Python CDISC programming pipeline — and I built them to be reviewed, with held-out benchmarks that report where the system is weak. I'm not asking Promptly to bet on potential. I'm

asking for a harder problem and better reviewers."

### If he probes the risk: "Are you sure you want to stay technical?"

"Yes — and specifically on the harmonisation and mapping side, not as a route into management. The thing I enjoy is the boundary where clinical meaning meets engineering discipline. What I want from Promptly is scale: I've built against synthetic data and one hospital-CSV contract. I want to see forty hospital data models and find out which of my assumptions survive."

---

## 2.3 — "You're doing the IBM Data Science Certificate. Are you trying to move away from data engineering?"

### What he is testing

**Commitment dilution.** He has seen it: someone joins as a data engineer, spends six months on ML courses, then asks to be reassigned to modelling. He wants to know whether your core craft is the pipeline.

### The reframe that wins

**Do not defend the certificate. Re-cast it as a translation layer for the same job.**

"The opposite — I'm doing it because the pipeline work keeps producing questions I can't currently answer, and I don't like shipping artefacts I can't defend.

Let me give you the concrete example. My held-out benchmark reports **accepted precision, coverage, mappable recall and abstain accuracy** on forty cases. That's a small sample. I can report those numbers honestly, but I can't yet put a **confidence interval** on them, and I can't yet say whether a 0.90 selection threshold is statistically distinguishable from 0.85 on that sample size. Right now the honest answer is 'I don't know, and the benchmark says so.'

That matters commercially. When Max asks a pharma client 'can the network answer this?', the answer includes 'how confident are we in the mappings underneath it.' I'd rather be able to quantify that than assert it.

**But the certificate is a second instrument, not a new career.** My primary craft is deterministic, auditable, reproducible ETL — and I'm protective of it. The applied-statistics content makes me better at evaluating what my pipeline produces. It doesn't make me want to build models instead of pipelines. If anything it's made me more conservative: the more I learn about calibration, the less I trust an LLM's self-reported confidence score, which is precisely why my governed score is the conservative `min(retrieval_score, llm_confidence)` rather than the model's own number.

Concretely: I'd like to grow toward being able to **read and critically appraise** an OHDSI population-level estimation study, and eventually run one. I don't need to be the statistician. I need to be the person who can tell the statistician whether the data underneath can support the design — and that's a data engineering skill with statistics literacy attached."

### The line to memorise

**"I'm not moving from pipelines to models. I'm moving from building pipelines to being able to prove what they produce."**

---

# 3. PITFALLS & BLIND SPOTS

## 3.1 — The traps a senior data lead sets for juniors

These are the questions designed to separate "read the OMOP spec" from "has been burned by real data."

| # | The trap | The junior answer that fails | The answer that passes |
|---|---|---|---|
| 1 | **"What's your mapping accuracy?"** | "95%." | "I don't publish a run-independent accuracy number, deliberately. My repo states that coverage, precision and recall must be generated for a specific `run_id`, because a percentage without a run, a vocabulary version and a split is not evidence." |
| 2 | **"Where do you put a lab result with no LOINC code?"** | Drop it, or map to a parent. | "Measurement with `concept_id = 0`, source value, source concept and unit provenance preserved. Declaration of unknown, not deletion." |
| 3 | **"Two source codes map to the same standard concept. Problem?"** | "No, that's fine." | "It's expected, and I check *why*. If it's granularity collapse, that's a real loss — reverberates into drug eras, where identical ingredients from different products should merge but different strengths must not. If it's genuine synonymy, fine. I need the lineage to know which." |
| 4 | **"Patient has a condition record dated after death."** | Clean it. | "First ask whether it's a data error or a registry artefact — a death record back-dated from a claims file, or a condition recorded at a post-mortem encounter. DQD plausibility flags it; the fix is a decision, not a deletion, and it must be logged because it changes the observation period." |
| 5 | **"Same drug, two brand names. One drug era or two?"** | Two. | "One, if they share an ingredient. I expand through `CONCEPT_ANCESTOR` to every current Standard Ingredient *before* deriving `DRUG_ERA`. But I keep `drug_source_value`, so a reviewer can see the products collapsed. Era derivation is lossy by design — which is exactly why lineage is non-negotiable." |
| 6 | **"How do you handle a concept valid until 2019 but your data is from 2023?"** | Use it anyway. | "I filter to current Standard concepts at retrieval: `standard_concept = 'S'`, no `invalid_reason`, and `CURRENT_DATE` between valid start and end. An expired mapping is a silent wrong answer, and expired concepts are one of the most common ways a pipeline that passed in 2022 quietly breaks in 2026." |
| 7 | **"What if a measurement belongs in Observation?"** | Guess the more common one. | "Domain routing comes from the concept's own `domain_id`, not from my intuition about the table. Where Condition and Observation disagree I route unresolved events once, retain unsupported routes in auditable quarantine, and let review decide. **Silent dropping is the unforgivable option.**" |
| 8 | **"How do you know your ETL didn't duplicate patients?"** | Row counts match. | "Row counts prove nothing about identity. I check deterministic 64-bit ID collision resistance in tests, and I bind identity through a governed pseudonymization-key fingerprint that fails the run if it changes. Duplication in a federated aggregate is invisible at the coordination centre — it just inflates your denominator." |
| 9 | **"Is your pipeline validated?"** | "Yes, tests pass." | "No. It's a portfolio reference implementation on synthetic data, with explicit limits documented. Tests establish behaviour under stated conditions, not fitness for regulated use. **Say this before he has to ask you.**" |
| 10 | **"What's the biggest weakness in your architecture?"** | (Hesitation / deflection) | "Retrieval recall. On held-out, Measurement top-k recall was 0.33 and overall 0.57. Reranking can't fix recall. That's where I'd invest first." |

> **The meta-trap:** trap #1 and trap #9 are the same trap. If you overclaim anywhere, João — who read your README's "Limits" row and "What are the limits?" table — will know you don't believe your own documentation. **Your documentation's honesty is your strongest asset. Do not contradict it in the room.**

---

## 3.2 — Where your pipeline would struggle on real Portuguese hospital EHR data

Have this list ready and volunteer it. Enumerating your own failure modes before being asked is the single most senior thing you can do in a junior interview.

**1. Language. This is the biggest one, and it's structural.**
Your embedding model is Chroma's default `all-MiniLM-L6-v2`, which is English-centric. Your LIS alias lexicon is English. Your LOINC axis regexes are English. Your prompt template is English. Meanwhile your own benchmark fixture already contains Portuguese cases — `"hipertensão arterial essencial"`, `"glicose 104, espécime e método ausentes"`, `"Hb sangue 13,6 g/dL"`, `"remoção cirúrgica do apêndice"`. Real consequence: Portuguese source text will retrieve worse than English source text, and that gap is currently **unmeasured** — your held-out set mixes languages without reporting per-language recall. What you'd do: **measure per-language recall first**, then evaluate a multilingual embedding model, and add a versioned `loinc-axes-v2-pt`. Not before the measurement — otherwise you're optimising blind.

**2. The LIS alias lexicon is deliberately inert on PHI.**
`scope: "generic-synthetic-development"` means the moment `data_classification` is anything other than `SYNTHETIC`, all eight aliases stop expanding. That's correct safety engineering and it means **your abbreviation handling effectively doesn't exist on real hospital data.** A real deployment needs an institution-authored lexicon with the same hash-pinning and the same fail-closed scope check. Be explicit that this is a designed gap, not an oversight.

**3. Real hospital data is mostly HL7 v2, not FHIR R4.**
Synthea emits clean FHIR. Portuguese hospitals run HL7 v2 ADT/ORU feeds, plus LIS and RIS CSV dumps, plus free-text. Your `hospital-csv-v1` adapter is the right shape of answer — a versioned, allowlisted, redacted contract producing **non-publishable pre-ingestion proposals** requiring later event binding — but it's one contract, not a general HL7 v2 parser. Expect a long tail of segment-level edge cases.

**4. Local codes without a registered source vocabulary.**
Your `SOURCE_IDENTITY_REGISTRY` boundary requires canonical system/code claims resolvable to one explicitly registered local OMOP source vocabulary. Real hospitals have coding systems that are undocumented, duplicated, or reused across domains with different meanings. Your architecture **fails closed here — which is right** — but it means coverage will start low at a new site and someone has to do the registration work.

**5. Scale and the LLM wall-clock.**
Your benchmark: **30 LLM calls took 82.8 seconds of model time** on a 7B local model, against a vocabulary of 68k Measurement / 157k Drug concepts. A real hospital has tens of thousands of distinct unmapped terms. Index preparation alone took **928 seconds**. Throughput is a genuine engineering problem, not a footnote — and it argues strongly for deterministic-first, LLM-last, with reuse policy shrinking the LLM surface every run.

**6. Two-reviewer governance needs real clinical capacity.**
Two blinded reviews plus separate adjudication is correct and it is **expensive in clinician time.** At a new site with thousands of novel terms, that's a resourcing conversation, not just a design one. Your mitigation — rejection as reusable policy, so the model isn't called again — is the right lever, and it's worth naming to Max.

**7. PHI reality.**
Your privacy boundary is well designed: loopback-only Ollama, direct-identifier redaction before prompting, metadata-only audit logs, role allowlists, and an explicit statement that the Streamlit portal **is not an identity provider.** In a Portuguese hospital deployment, identity, retention and access control are institution-managed. **Never imply your tooling solves that.**

**8. Units and reference ranges.**
`unit_mapping.py` handles UCUM correctly with three explicit outcomes (known → Standard `Unit` concept; supplied-but-unsupported → `0`; genuinely absent → `NULL`). Real LIS feeds add the harder cases: `mg/dL` vs `mmol/L` convention drift across sites, non-UCUM local unit strings, and ratios. This is where the

Biochemistry background is genuinely load-bearing — say so.

**9. Time and temporality.**
Synthea is temporally clean. Real EHRs have back-dated administrative records, unknown-time outpatient events, and transfers between institutions that create overlapping observation periods. Your `ADTTE` work shows you understand censoring; `OBSERVATION_PERIOD` derivation from DM reference dates shows you understand why naive derivation biases longitudinal analysis. **Bring that up on your own — it's the bridge between TTE and ETL that most candidates miss.**

---

# 4. WINNING REVERSE-QUESTIONS

Ask them at the end. All three are grounded in Promptly's own published positions, and none can be answered by someone who only skimmed your README.

---

### Question 1 — for João (the harmonisation bottleneck)

> **"In your OHDSI poster on the AI-based mapping platform, the loop is: the AI suggests, clinicians review, and the finalised mappings are exported into the database. In your interview you described USAGI plus clinical specialists, and said the goal is to retrain on the source database's concepts to reduce manual work.**
>
> **What I can't tell from outside is what happens to a disagreement. When two clinical specialists disagree on a target concept, or when a specialist overrides the AI — does that resolution get captured as structured, reusable signal, or does it end up as a corrected row with the reasoning lost?**
>
> **I ask because that's the design decision I spent most of my time on. My rejections become active policy that suppresses identical future proposals and prevents the model call entirely — so human effort decays as the mapping set matures. And I found that separating the decision from the rationale was the hard part: the rationale is what makes the next decision cheaper. If you're capturing that, there's an obvious next step where reviewer disagreement becomes a training or calibration signal. If you're not, that's arguably the highest-value thing to instrument — because right now the most expensive, highest-signal moments in your pipeline are the ones leaving the least trace."**

Why it works: you have read their **poster and their interview**, you understand **reuse economics**, and you're offering a concrete improvement rather than a compliment. It also flatters João's published thesis without flattering him.

---

### Question 2 — for João (the federated quality problem)

> **"In a centralised database, DQD is a gate I control. In your network, each partner runs the same harmonisation container, but the source differs — different LIS vendors, different coding conventions, different completeness. Two partners can both publish a v5.4 CDM that passes DQD, and still be non-comparable, because DQD only measures internal coherence against the CDM contract, never whether a mapping is clinically right.**
>
> **So: how do you make quality comparable across nodes? Is there a shared, network-level allowance register — a governed list of 'this check fails at every Portuguese hospital because the LIS never captured the field' — or does each partner maintain its own, which would make cross-site feasibility queries very hard to interpret?**
>
> **And the thing I'd genuinely want to know: when a study runs across, say, six nodes and one node is an outlier, how do you distinguish a real epidemiological difference from a**

> **harmonisation artefact? Because in a federated model you can't go look at the rows."**

Why it works: this is a **real, unsolved, federated-specific** problem, it's exactly João's job, and your own bidirectional stale-allowance check shows you've already hit the single-site version of it. You are asking to be handed the multi-site version.

---

### Question 3 — for Max (commercial and strategic)

> **"FRAME is emulating antihypertensive treatment in pregnancy across multiple TREs, and asking whether effects vary by ethnicity. That's a study where two of the hardest variables are the ones observational data handles worst: ethnicity is inconsistently recorded and inconsistently coded across jurisdictions, and the exposure is pregnancy — a time-dependent, non-static state.**
>
> **My question is about the commercial sequence. When a pharma client comes to you, do they arrive with a protocol and ask 'can the network run this?' — in which case your feasibility answer has to be honest about the fact that DQD passing doesn't mean the phenotype is right — or do you increasingly shape the protocol with them, using what you know about what the network can actually observe?**
>
> **Because the thing I built that I think is most relevant here isn't the LLM — it's the lineage. In my CDISC work, every OMOP row traces to an exact SDTM record, and every mapping decision carries the run, model digest, vocabulary version and the two reviewers and adjudicator behind it. In an HTA or regulatory submission, 'we mapped it' isn't an answer; 'here is exactly where this came from' is. I'd want to know whether that provenance story is something your clients are already asking for — or whether it's the thing you'd want to build next."**

Why it works: you named **FRAME correctly** (federated antihypertensives-in-pregnancy emulation across TREs, ethnicity effect modification — from DARE UK's public description), you identified the two variables that make it hard, and you connected it to **HTA acceptance**, which is Max's actual commercial anxiety. Then you offered your differentiator as a question rather than a pitch.

---

## Backup questions (if time allows)

- **To Max:** "Explore's POLLY is about agreeing what 'a readmission' or 'uncontrolled diabetes' means before anyone queries — Define, Approve, Trace. That's structurally the same problem as vocabulary mapping: multiple plausible standard concepts, and the choice changes the result. Is POLLY's definition library stored in a way that's reusable across partners, or is each partner's library local? I ask because cross-site reuse of definitions might be a bigger lever than cross-site reuse of mappings."
- **To João:** "Where does the network sit on EHDS — are you positioning as a data-holder-side harmonisation layer for HDABs, or building toward HealthData@EU federation directly? I built my pipeline to produce a hashable, reproducible artefact per run precisely because that seemed like the unit a regulator would want to audit, but I'd love to know whether that instinct matches where the regulation is landing."
- **To both:** "What does the first ninety days look like for someone in this role — is the first project a new-partner onboarding, or hardening an existing harmonisation path?" (Signals you intend to start immediately.)

---

# APPENDIX — ONE-PAGE CHEAT SHEET

## Numbers to have at your fingertips

| Metric | Value | Context |
|---|---|---|
| Held-out benchmark | **40 cases** (30 MAP / 10 ABSTAIN) | `dirty-hospital-to-omop`, split `held_out` |
| Accepted precision | **1.00** | Both deterministic and LLM arms |
| Coverage | **0.25** | Both arms — identical |
| False maps | **0** | Both arms |
| Abstain accuracy | **1.00** | |
| Threshold 0.0 (embedding arm) | precision **0.35**, coverage **1.00**, **10 false maps** | The cost of "100% coverage" |
| Top-k recall, overall | **0.567** (17/30) | **Your weakest number — volunteer it** |
| Top-k recall, Measurement | **0.333** (2/6) | Worst domain |
| Top-k recall, Condition | **0.833** (5/6) | Best domain |
| Model | `qwen2.5-coder:7b`, temp **0.0**, seed **0** | 30 calls, **82.8 s** model time, **0 contract failures** |
| Prompt version | `mapping-json-v2` (current runtime: `mapping-json-v5`) | Know both; say which is which |
| LOINC pool / top-k | **20 → 5** | `loinc-axes-v1` |
| Axis scoring | **+0.08 match / −0.35 conflict** | Asymmetry is the design |
| DQD policy | `max_errors: 0`, `max_failed_checks: 0` | Bidirectional allowance enforcement |
| DQD stack | DQD **2.8.9**, DatabaseConnector **7.2.0**, SqlRender **1.19.6** | CDM **5.4** |
| Index prep | **928 s**; Measurement **68,106** concepts | Drug 157,190 · Observation 129,490 · Condition 105,324 · Procedure 48,344 |
| Vocabulary | **v5.0 27-FEB-26** | DuckDB **1.5.5**, Python **3.12.10** |
| Tests | **56 files**, **345 test functions**, **37 parametrize decorators** | [!] **Run `pytest --collect-only -q` and write down the exact collected number before Thursday.** |

**Curation status to state verbatim:** `PROVISIONAL_TECHNICAL`, `clinical_validation_required: true`.

## Things you must never say

- [X] "Around 95% mapping accuracy." → Your repo explicitly refuses to publish run-independent percentages.
- [X] "It's validated." → It's a portfolio reference implementation on synthetic data with documented limits.
- [X] "The LLM picks the right concept." → The LLM proposes. Two humans and an adjudicator decide.
- [X] "DuckDB scales better than Spark." → Say "the transform isn't shuffle-bound; the serving layer is a different decision."
- [X] "We handle PHI securely." → Identity, retention and access are institution-managed. Your tooling is a boundary, not a solution.
- [X] Exact Promptly network figures from memory. → Their own pages say +70M lives / 250 hospitals / 35M lives / 400 hospitals / 100M+ lives depending on date and page. **Say ranges: "tens of millions of lives across hundreds of hospitals in six-plus countries."** Getting a stale number wrong in front of Max is worse than being vague.
- [X] "I'm transitioning away from support work." → Reframe as stakeholder empathy + operational precision.

## Facts about Promptly worth having ready

- **Federated**: "moving insights, not data" — data never leaves the partner's servers; the Data Partner remains data controller under GDPR.
- **Promptly Harmonize**: OMOP CDM **v5.4**; Mapping Platform System; quality dashboards and metadata catalogues; "research-ready datasets in weeks, not months."
- **Explore by Promptly**: natural-language and live cohort discovery; **POLLY** — governed library of clinical/operational definitions, Define → Approve → Trace.
- **OMOP-CDM Certified** by the European Health Data & Evidence Network (EHDEN).
- **Datavant partnership** (Apr 2024) — tokenisation layer.
- **TrustChain** (Dec 2024) — federated learning governance gateway with IPN, ULS Coimbra, ULS Santo António.
- **ELSAN** (Sept 2025) — French entry; **Hapvida NotreDame Intermédica** (June 2026) — Brazil, ~16M beneficiaries.
- **FRAME** — DARE UK Research Exemplars, £446,795: Federated Research on Anti-hypertensives Maternity Emulation Trial, federated analytics across multiple TREs, effect modification by ethnicity, child developmental outcomes.
- **EHDS** — Regulation adopted; OMOP + FHIR as the de facto "dual engines"; DARWIN EU as the pathfinder.

## Your 90-second opening (practise until it's boring)

> "Thanks for having me. Quick frame on where I'm coming from.
>
> I'm a biochemist who moved into clinical data engineering, and I've spent the last while building the thing I think is the hard part of RWE: **getting heterogeneous clinical data into OMOP without letting an automated step make an invisible decision.**
>
> The main project is `FHIR-to-OMOP` — Synthea FHIR R4 into OMOP CDM v5.4. Three pieces I'd point you at. **First**, deterministic-first mapping: source codes, vocabulary and `Maps To` relationships run before any AI touches the data, with deterministic tie-breaking to prevent fan-out. **Second**, for the messy residue — legacy LIS abbreviations, mixed-language text — retrieval plus a local LLM, but the model can only select from candidates it was actually shown, must return strict JSON, and may abstain. Invalid output fails closed. **Third**, and the part I care most about: **an LLM output is never a mapping.** It's a proposal with a run ID, model digest, prompt version, vocabulary version and retrieval index signature attached, and it needs two independent blinded reviews plus separate adjudication before it can enter a clinical table. Rejections become reusable policy, so the model isn't even called next time.
>
> I validate the output with OHDSI DQD, run as a **gate**, not a dashboard — zero error budget, and an allowance register that's enforced in both directions, so a stale exemption fails the build.
>
> And the number I'd lead with is the unflattering one: **on my held-out set, retrieval top-k recall was 0.57 overall and 0.33 for Measurement.** The LLM arm matched the deterministic arm on precision and coverage rather than beating it. Reranking can't fix recall, so that's where I'd invest next.
>
> I also built `CDISC-to-OMOP` with record-level SDTM lineage, and a SAS/Python CDISC programming pipeline — which is where I got interested in target trial emulation, because time-zero alignment and observation-period derivation are ETL decisions, not statistical ones.
>
> Happy to go as deep as you want on any of it."

## The day before

- [ ] Run `pytest --collect-only -q` — write the exact collected count on your hand.

- [ ] Re-read `src/mapping/loinc_reranking.py` (156 lines) and `src/clinical_mapping_core/decision.py` (153 lines). You will be asked about them line by line.
- [ ] Re-read `quality/dqd_policy.json` and `src/quality/validate_dqd.py`. Know why stale allowances fail the build.
- [ ] Say the 90-second opener out loud five times. Out loud, not in your head.
- [ ] Practise saying **"I don't know, and here's how I'd find out"** in a calm voice. You will use it at least twice.
- [ ] Have one question ready for each of them written on paper in front of you. Asking it is better than remembering it.

---

Prepared from a full read of the three repositories and Promptly Health's published technical positions. Every number in this document is traceable to a file in `FHIR-to-OMOP` or a public Promptly/DARE UK source. Where a figure is a limitation, it is labelled as one — that is deliberate, and it is your strongest asset in the room.